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United States Patent	12400909
Kind Code	B2
Date of Patent	August 26, 2025
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Method for producing support substrate for bonded wafer, and support substrate for bonded wafer

Abstract

A producing method of a handle wafer for a bonded wafer produced by bonding an active wafer and the handle wafer through an insulation film includes: preparing a handle wafer body made from a monocrystalline silicon wafer; forming an oxide film on the handle wafer body; depositing a polycrystalline silicon layer on the oxide film; forming a protective oxide film on a surface of the polycrystalline silicon layer; and polishing to remove the protective oxide film and polishing the polycrystalline silicon layer.

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Appl. No.:	18/032247
Filed (or PCT Filed):	October 18, 2021
PCT No.:	PCT/JP2021/038375
PCT Pub. No.:	WO2022/091831
PCT Pub. Date:	May 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230395423 A1	Dec. 07, 2023

Foreign Application Priority Data

JP 2020-179040 Oct. 26, 2020

Publication Classification

Int. Cl.: H01L21/304 (20060101); C23C16/24 (20060101); C23C16/56 (20060101);
H01L21/762 (20060101)

U.S. Cl.:

CPC H01L21/76254 (20130101); C23C16/24 (20130101); C23C16/56 (20130101);

Field of Classification Search

CPC: H01L (21/304)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a producing method of a handle wafer for a bonded wafer and a handle wafer for a bonded wafer.

BACKGROUND ART

(2) Silicon On Insulator (SOI) wafers have been used as substrates for high-frequency (Radio Frequency (RF)) devices. The SOI wafers each include an insulation film made of, for instance, silicon oxide (SiO_2) and an active layer (e.g., monocrystalline silicon), which are sequentially layered on a handle wafer (e.g., a monocrystalline silicon wafer).

(3) One of typical methods for producing the SOI wafers is a bonding method. In the bonding method, the insulation film is formed on at least one of the handle wafer or the active wafer, then these wafers are bonded through the insulation film and subsequently subjected to heat treatment at a high temperature of approximately 1,200 degrees C. to produce the SOI wafer (hereinafter, the SOI wafer produced by the bonding method is referred to as a “bonded wafer”).

(4) In order to address RF in the bonded wafer, the handle wafer has been increased in resistivity (for instance, to have a resistivity of 3000 $\Omega \cdot \text{cm}$ or more). However, there is a demand for bonded wafers handling higher frequency in response to the need for higher-speed devices and this demand cannot be met only by increasing the resistivity of the handle wafer.

(5) In view of the above, it has been proposed to provide, on a surface of the handle wafer, a polycrystalline silicon layer as a carrier trap layer that traps and annihilates carriers generated during a high-frequency operation (see, for instance, Patent Literature 1). In this method, in order to prevent epitaxial growth of silicon on the monocrystalline silicon of the handle wafer, an ultra-thin oxide film is formed on the handle wafer and polycrystalline silicon is formed on the ultra-thin oxide film. A surface provided with the polycrystalline silicon is polished and then bonded to an insulation film formed on the active layer.

CITATION LIST

Patent Literature(s)

(6) Patent Literature 1: JP 2000-200741 A

SUMMARY OF THE INVENTION

Problem(s) to be Solved by the Invention

(7) Here, the generation of void defects in the bonding step disadvantageously causes film peeling in the bonded wafer to decrease the yield in bonding. In this regard, Patent Literature 1 describes the technique of controlling a surface roughness of the polycrystalline silicon layer to be 1 nm or less in terms of a centerline average roughness R_a to inhibit the generation of void defects in heating for bonding, thereby attaining a required strength.

(8) However, even in a case where the bonded wafers are produced through the above technique, void defects may be generated. Thus, further measures are desirably taken to improve the quality of the bonded wafers.

(9) An object of the invention is to provide a producing method of a handle wafer for a bonded wafer, the handle wafer capable of inhibiting void defects in a bonding step, and to provide such a handle wafer for a bonded wafer.

Means for Solving the Problem(s)

(10) According to an aspect of the invention, a producing method of a handle wafer for a bonded wafer produced by bonding an active wafer and the handle wafer through an insulation film includes: preparing a handle wafer body made from a monocrystalline silicon wafer; forming an oxide film on the handle wafer body; depositing a polycrystalline silicon layer on the oxide film; forming a protective oxide film on a surface of the polycrystalline silicon layer; and polishing to

remove the protective oxide film and polishing the polycrystalline silicon layer.

(11) In the production method of a handle wafer for a bonded wafer according to the above aspect of the invention, in the forming the protective oxide film, the protective oxide film may be formed through SC-1 cleaning.

(12) In the production method of a handle wafer for a bonded wafer according to the above aspect of the invention, in the forming the protective oxide film, the protective oxide film may be formed through ozone passivation.

(13) In the production method of a handle wafer for a bonded wafer according to the above aspect of the invention, the protective oxide film formed in the forming the protective oxide film may have a thickness in a range from 0.5 nm to 10 nm.

(14) According to another aspect of the invention, a handle wafer for a bonded wafer produced by bonding an active wafer and the handle wafer through an insulation film includes: a handle wafer body; and a polycrystalline silicon layer deposited on the handle wafer body, in which a root-mean-square roughness R_q of a polished surface of the polycrystalline silicon layer, which is measured in an area of $10\text{ }\mu\text{m}\times 10\text{ }\mu\text{m}$, is 0.5 nm or less, the polycrystalline silicon layer has a thickness in a range from $1.5\text{ }\mu\text{m}$ to $2.0\text{ }\mu\text{m}$, and the number of pits of 2 nm or more detected in the surface of the polycrystalline silicon layer in a DIC mode of SP-1 produced by KLA-Tencor Corporation is 1 piece/ cm^2 or less.

(15) According to the above aspects of the invention, the generation of void defects in the bonding step can be inhibited with the handle wafer used for the bonded wafer produced by bonding the active wafer and the handle wafer through the insulation film.

Description

BRIEF EXPLANATION OF DRAWINGS

(1) FIG. 1 is a flowchart for illustrating a process for producing a bonded wafer according to an exemplary embodiment of the invention.

(2) FIG. 2 is a cross-sectional view showing steps of a production method of the bonded wafer according to the exemplary embodiment of the invention.

DESCRIPTION OF EMBODIMENT(S)

(3) Suitable exemplary embodiment(s) of the invention will be described in detail below with reference to the attached drawings. A handle wafer for a bonded wafer of the invention is, for instance, used for a bonded wafer (e.g., an SOI wafer), which is obtained by bonding the handle wafer to an insulation film formed on an active wafer.

(4) Inventors of the invention have made dedicated studies on the handle wafer for the bonded wafer, the handle wafer capable of inhibiting the generation of void defects. As a result, it has been found that in polishing a polycrystalline silicon layer of the handle wafer, for instance, impurities (e.g., slurry residues) caused by a polishing apparatus adhere on the polycrystalline silicon layer and these impurities cause defects that are to be detected.

(5) Specifically, it has been found that by polishing the polycrystalline silicon layer on which a minute amount of slurry residues adheres, a surface of the polycrystalline silicon layer is locally etched to generate pits (i.e., pit defects), so that these pits cause defects after the handle wafer is bonded to the active wafer. A production method of the handle wafer of the invention includes forming a protective oxide film for protecting the polycrystalline silicon layer in order to eliminate the effect of the impurities.

(6) FIG. 1 is a flowchart for illustrating a process for producing the bonded wafer of the exemplary embodiment. FIG. 2 is a cross-sectional view showing steps of the production method of the bonded wafer.

(7) As shown in FIG. 1, the production method of the bonded wafer includes an active wafer

production step **S1** for producing the active wafer, a handle wafer production step **S2** for producing the handle wafer (i.e., the production method of the handle wafer), which is independent of the active wafer production step **S1**, and a bonded wafer production step **S3** for producing the bonded wafer by bonding the active wafer and the handle wafer.

(8) The active wafer production step **S1** includes an active wafer body preparation step **S11**, an insulation film growth step **S12**, an ion implantation layer formation step **S13**, and a pre-bonding cleaning step **S14**.

(9) In the active wafer body preparation step **S11**, an active wafer body **10** in a form of a monocrystalline silicon wafer is prepared as shown in FIG. 2(a). In the insulation film growth step **S12**, an insulation film **11** (i.e., an oxide film) is grown around the active wafer body **10** through, for instance, thermal oxidation, CVD or the like as shown in FIG. 2(b).

(10) In the ion implantation layer formation step **S13**, an ion implantation layer **12** is formed in the active wafer body **10** by implanting hydrogen ion or rare gas ion from above the insulation film **11** using an ion implanter.

(11) In the pre-bonding cleaning step **S14**, the active layer is cleaned before being bonded in order to remove particles on a bonding surface of the active wafer body **10**.

(12) The active wafer **13** for the bonded wafer is produced through the above steps.

(13) The handle wafer production step **S2** includes a handle wafer body preparation step **S21**, an oxide film formation step **S22**, a polycrystalline silicon layer deposition step **S23**, a protective oxide film formation step **S24**, a polishing step **S25**, and a pre-bonding cleaning step **S26**.

(14) In the handle wafer body preparation step **S21**, a handle wafer body **20** made from a monocrystalline silicon wafer is prepared as shown in FIG. 2(d). The monocrystalline silicon wafer to be used may be produced by slicing a monocrystalline silicon ingot, which is grown by a Czochralski method (CZ method) or floating zone melting method (FZ method), using a wire saw or the like.

(15) In the oxide film formation step **S22**, an oxide film **21** is formed on a bonding surface of the handle wafer body **20** as shown in FIG. 2(e). Preferably, the thickness of the oxide film **21** is, for instance, in a range from 0.3 nm to 10 nm. With the reduced thickness of the oxide film **21**, the oxide film **21** interposed between the handle wafer body **20** and a later-described polycrystalline silicon layer **22** hardly affects properties of the RF device.

(16) The oxide film **21** may be formed through, for instance, wet cleaning such as alkali cleaning (SC-1 cleaning) or acid cleaning (SC-2 cleaning). The oxide film **21** may be formed by any other method, for example, by thermal oxidation under an oxidative atmosphere or oxidation heat treatment using a rapid heating/cooling device.

(17) In the polycrystalline silicon layer deposition step **S23**, a polycrystalline silicon layer **22** is deposited on the oxide film **21** formed on the bonding surface of the handle wafer body **20** as shown in FIG. 2(f). The polycrystalline silicon layer **22** may be deposited through, for instance, a CVD method. Preferably, the thickness of the polycrystalline silicon layer **22** is in a range from 2 μm to 4 μm .

(18) The protective oxide film formation step **S24** is a step of forming, on a surface of the polycrystalline silicon layer **22**, a protective oxide film **23** serving as a protective film, as shown in FIG. 2(g). Preferably, the thickness of the protective oxide film **23** is, for instance, in a range from 0.5 nm to 10 nm. With the thickness of the protective oxide film **23** being less than 0.5 nm, the protective oxide film **23** cannot exhibit the effect as a protective film. With the thickness of the protective oxide film **23** being more than 10 nm, time required for polishing in the polishing step **S25** increases, leading to a high production cost.

(19) In order to attain a balance between the effect as a protective film and the production cost, the thickness of the protective oxide film **23** is more preferably in a range from 0.7 nm to 2 nm.

(20) The protective oxide film **23** is formed through alkali cleaning (SC-1 cleaning). Specifically, in the protective oxide film formation step **S24**, an oxide film is formed on the surface of the

polycrystalline silicon layer **22** by cleaning the surface of the polycrystalline silicon layer **22** with an aqueous solution of NH_4OH (ammonium hydroxide) and H_2O_2 (hydrogen peroxide).

(21) In the polishing step **S25**, the protective oxide film **23** is removed by being polished and the surface of the polycrystalline silicon layer **22** is polished to be flat as shown in FIG. **2(h)**. As a polishing method, a known method such as chemical mechanical polishing (CMP) is suitably usable.

(22) In the polishing step **S25**, the polishing is performed so that the thickness of the polycrystalline silicon layer **22** is in a range from 1.5 μm to 2.0 μm . Furthermore, in the polishing step **S25**, the polishing is performed so that a root-mean-square roughness R_q of the polished polycrystalline silicon layer **22**, which is measured in an area of $10\ \mu\text{m} \times 10\ \mu\text{m}$, is 0.5 nm or less. The thickness of the polycrystalline silicon layer **22** is measured at nine points within a plane of the handle wafer. The nine measurement points are a center point of the handle wafer; vertices of a square inscribed in a circle that is concentric with the handle wafer and has a radius half of that of the handle wafer; and vertices of a square inscribed in a circle that is concentric with the handle wafer and is located 6 mm inside an outer circumferential edge of the handle wafer.

(23) Here, in performing the polishing step **S25**, since the polycrystalline silicon layer **22** is covered by the protective oxide film **23**, impurities (e.g., slurry residues) caused by the polishing apparatus do not adhere on the polycrystalline silicon layer **22**. Thus, the polycrystalline silicon layer **22** is not eroded by the impurities.

(24) In the pre-bonding cleaning step **S26**, particles on the polished surface of the polycrystalline silicon layer **22** are removed.

(25) The handle wafer **24** for the bonded wafer is produced through the above steps. It should be noted that the active wafer production step **S1** (**S11** to **S14**) and the handle wafer production step **S2** (**S21** to **S26**) may be performed in parallel.

(26) Next, the bonded wafer production step **S3** for bonding the active wafer **13** and the handle wafer **24** to produce the bonded wafer will be described below.

(27) The bonded wafer production step **S3** includes a bonding step **S31**, a peeling heat treatment step **S32**, and a bonding heat treatment step **S33**.

(28) In the bonding step **S31**, the polished surface of the polycrystalline silicon layer **22** of the handle wafer **24** and the active wafer **13** are bonded through the insulation film **11** as shown in FIG. **2(i)**. At this time, the active wafer **13** and the handle wafer **24** are bonded so that an implantation surface of the active wafer **13** faces the polycrystalline silicon layer **22**.

(29) In the peeling heat treatment step **S32**, the ion implantation layer **12** is subjected to heat treatment (peeling heat treatment) for generating microbubbles to peel the ion implantation layer **12** with the generated microbubbles. The bonded wafer **30**, in which the insulation film **11** and an active layer **31** are formed on the handle wafer **24**, is thus produced as shown in FIG. **2(j)**. It should be noted that a peeled wafer **40** having a peeling surface **41** is also produced.

(30) In the bonding heat treatment step **S33**, the bonded wafer **30** is subjected to bonding heat treatment to enhance the bonding strength at the bonding interface.

(31) The bonded wafer **30** can be produced as described above.

(32) In the handle wafer **24** used in the above production method of the bonded wafer, the root-mean-square roughness R_q of the polished surface of the polycrystalline silicon layer **22**, which is measured in an area of $10\ \mu\text{m} \times 10\ \mu\text{m}$, is nm or less. Further, the number of pits of 2 nm or more in the surface of the polycrystalline silicon layer **22** of the handle wafer **24** is 1 piece/cm² or less. The pits are detected in the DIC mode of SP-1 produced by KLA-Tencor Corporation.

(33) According to the handle wafer production step **S2** of the above exemplary embodiment, the polycrystalline silicon layer **22** is not eroded by impurities in performing the polishing step **S25**. Accordingly, the formation of pits caused by the impurities can be inhibited and thus the generation of void defects due to the pits after the handle wafer is bonded to the active wafer can be inhibited.

(34) In the protective oxide film formation step S24 of the above exemplary embodiment, the protective oxide film 23 is formed through SC-1 cleaning but may be formed through any other method. For instance, the protective oxide film 23 may be formed through ozone passivation.

(35) In the protective oxide film formation step S24 using ozone passivation, the deposited polycrystalline silicon layer 22 is exposed to high-density ozone at normal temperature without being subjected to heat treatment, thereby forming the protective oxide film 23 on the surface of the polycrystalline silicon layer 22. By this operation, the dense protective oxide film 23 can be formed on the polycrystalline silicon layer 22.

(36) Further, a method of forming the protective oxide film 23 in the protective oxide film formation step S24 is not limited as long as the thickness of the protective oxide film 23 falls within a range from 0.5 nm to 10 nm. Specifically, the protective oxide film 23 may be a native oxide film, thermal oxide film, deposited oxide film, or the like.

EXAMPLES

(37) Next, the invention will be described in more detail below with reference to Examples. However, it should be noted that the scope of the invention is by no means limited by these Examples.

(38) In order to examine the effect of the protective oxide film of the invention, a handle wafer in which a polycrystalline silicon layer was deposited and then polished without forming thereon a protective oxide film (Comparative) was compared with a handle wafer in which a polycrystalline silicon layer was deposited, then a protective oxide film was formed thereon and the resultant layer and film were polished (Examples 1 and 2), regarding the number of pits after the polishing.

Comparative

(39) A polycrystalline silicon layer was deposited on the handle wafer body and then polished without forming thereon a protective oxide film, thereby producing a handle wafer.

Example 1

(40) A polycrystalline silicon layer was deposited on the handle wafer body, then a protective oxide film was formed thereon through SC-1 cleaning and subsequently the resultant layer and film were polished, thereby producing a handle wafer.

Example 2

(41) A polycrystalline silicon layer was deposited on the handle wafer body, then a protective oxide film was formed thereon through ozone passivation and subsequently the resultant layer and film were polished, thereby producing a handle wafer.

(42) The handle wafers of Comparative and Examples 1 and 2 were produced by the same method except for the presence or absence of the protective oxide film and/or the method of forming the protective oxide film.

(43) Table 1 shows comparison among Comparative and Examples 1 and 2 regarding the average number of pits on the polycrystalline silicon layer. The number of pits refers to the number of pits of 2 nm or more detected in the DIC mode of SP-1 produced by KLA-Tencor Corporation. The measurement was performed on a plurality of handle wafers in each of Comparative and Examples 1 and 2. Table 1 shows the respective average numbers thereof.

(44) TABLE-US-00001

TABLE 1	Comparative	Example 1	Example 2	Average Number of Pits
	1.14	0.39	0.49	(pieces/cm.sup.2)

(45) As is clear from Table 1, Comparative has an average number of pits of 1.14 pieces/cm.sup.2, whereas Examples 1 and 2 both have an average number of pits of 1 piece/cm.sup.2 or less as Example 1 in which the protective oxide film was formed through SC-1 cleaning has an average number of pits of 0.39 pieces/cm.sup.2 and Example 2 in which the protective oxide film was formed through ozone passivation has an average number of pits of 0.49 pieces/cm.sup.2.

Claims

1. A producing method of a handle wafer for a bonded wafer produced by bonding an active wafer and the handle wafer through an insulation film, the method comprising: preparing a handle wafer body made from a monocrystalline silicon wafer; forming an oxide film on the handle wafer body; depositing a polycrystalline silicon layer on the oxide film; forming a protective oxide film on a surface of the polycrystalline silicon layer through SC-1 cleaning of the polycrystalline silicon layer; and polishing to remove the protective oxide film and polishing the polycrystalline silicon layer.
 2. The production method of a handle wafer for a bonded wafer according to claim 1, wherein the protective oxide film formed in the forming the protective oxide film has a thickness in a range from 0.5 nm to 10 nm.
 3. A producing method of a handle wafer for a bonded wafer produced by bonding an active wafer and the handle wafer through an insulation film, the method comprising: preparing a handle wafer body made from a monocrystalline silicon wafer; forming an oxide film on the handle wafer body; depositing a polycrystalline silicon layer on the oxide film; forming, on a surface of the polycrystalline silicon layer, a protective oxide film having a thickness in a range from 0.5 nm to 10 nm; and polishing to remove the protective oxide film and polishing the polycrystalline silicon layer.
 4. The production method of a handle wafer for a bonded wafer according to claim 3, wherein in the forming the protective oxide film, the protective oxide film is formed through SC-1 cleaning.
 5. The production method of a handle wafer for a bonded wafer according to claim 3, wherein in the forming the protective oxide film, the protective oxide film is formed through ozone passivation.
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